| | | |
|---|---|---|
| | $= 2.21 \times 10$ s $= 2.2 \times 10$ s | [1] answer |
| **8(a)(i)** | 20.8 knots = $10.6912\ \text{m s}^{-1}$<br><br>Maximum KE of ship<br>$=\frac{1}{2}\left(3.40\times10^{7}\right)\left(10.6912\right)^{2}$<br>$=1.94\times10^{9}\ \text{J}$ | [1] for correct substitution<br>[1] for correct answer |
| **8(a)(ii)** | Angular speed<br>$=\frac{115\left(2\pi\right)}{60}$<br>$=12.0428\ \text{rad s}^{-1}$<br>$\approx 12.0\ \text{rad s}^{-1}$ | [1] for correct substitution<br><br>[1] for correct answer |
| **8(b)** | Power required to convert water at 100 °C to steam at 100 °C every hour<br>$=\frac{\left(2.15\times10^{5}\right)\left(2.26\times10^{6}\right)}{3600}$<br>$=1.3497\times10^{8}\ \text{W}$<br><br>Maximum efficiency<br>$=\frac{29400\times10^{3}}{1.3497\times10^{8}}\times100\%$<br>$=21.8\%$ | [1] for correct substitution for power required<br><br>[1] for correct substitution for efficiency<br>[1] for correct answer |

| | | |
|---|---|---|
| **8(c)(i)** | In the case of fission, the neutrons produced by each reaction causes further neutron capture by other uranium atoms. This in turn produces more neutrons to cause further reactions. | [1] |
| **8(c)(ii)** | At the beginning, the number of neutrons available for capture is small but this escalates to a tremendous value towards the final stages, when there are a large number of simultaneous reactions taking place at the same time. | [1] |
| **8(c)(iii)** | Electromagnetic radiation.<br>One example is gamma radiation, light/photons. | [1] |
| **8(d)(i)** | 108, 127, 152<br><br>(All have a percentage yield of 0.06% to 0.08%.) | [2] for all correct<br>[1] for one or two correct |
| **8(d)(ii)** | By conservation of mass number<br><br>235 + 1 = 82 + c + 2<br>OR<br>235 + 1 = 82 + c + 3<br><br>Therefore c = 152 or 151<br>Based on **(d)(i)**, c = 152 | [1] |
| **8(d)(iii)** | 6.1% ≤ percentage yield ≤ 6.9% | [1] |
| **8(d)(iv)** | For fission products to have the same masses, the nucleon number must be around 0.5(236 – 2) = 117 (the subtraction of 2 takes into account the estimated number of neutrons produced).<br><br>For the nucleus with nucleon number 117, the percentage yield is 0.01%.<br><br>Taking the yield as 6.4%,<br>the ratio is 6.4% divided by 0.01% = 640 times. | [1]<br><br><br><br>[1] if use 0.012% incorrect |
| **8(e)** | $\beta^-$-particles are fast moving electrons and the $\beta^-$-particles emitted will collide with the atoms of the lead container.<br><br>These collisions result in the $\beta^-$-particles undergoing deceleration, either being slowed down by collision (or by deflection).<br><br>The electrons will then release energy in the form of the continuous spectrum of the X-ray produced, also called Bremsstrahlung radiation. | [1] for fast moving electrons<br>[1] for collision with atoms<br>[1] for deceleration of $\beta^-$-particles<br>[1] for releasing energy in the form of X-ray |